Supplementary Materials of "Niching-Based Ant Colony Optimization With Multiple Pheromone Matrices for Multi-objective Feature Selection in Classification"

TABLE S1
Key notations used in the proposed method.

Symbol	Meaning
$\eta_{i,j}$	Unified heuristic value for feature pair (i,j)
$\eta_{i,j,1}, \eta_{i,j,3}$	Heuristic value for excluding feature F_j
$\eta_{i,j,2}, \eta_{i,j,4}$	Heuristic value for selecting feature F_j
U_0^k	Unvisited features corresponding to the discarded branch
U_1^k	Unvisited features corresponding to the selected branch
$\tau_{i,j}^{m,k}$	Pheromone value on edge (i,j) for ant k in niche m

I. EXPERIMENTAL DESIGN

A. Parameter Setting

To further investigate the sensitivity of the three control parameters of NMACO to dataset dimensionality, the 17 benchmark datasets are divided into three groups: low-dimensional ($D \leq 100$), medium-dimensional ($100 < D \leq 500$), and high-dimensional ($D > 500$). For each group, the 25 parameter configurations generated by the Taguchi $L_{25}(5^3)$ orthogonal array are evaluated using the average HV performance. The complete results are reported in Tables S2, S3 and S4.

Based on the range analysis of the Taguchi orthogonal experiment results, NMACO exhibits varying parameter sensitivity across different dataset dimensionalities. In low-dimensional scenarios ($D \leq 100$), the algorithm demonstrates high robustness and low sensitivity, with most parameter configurations yielding stable mean HV values around 0.820 and configuration K16 achieving the top mean HV of 0.826. As dimensionality increases to medium-dimensional spaces ($100 < D \leq 500$), parameter sensitivity becomes significantly more pronounced; improper settings such as K1 lead to noticeable performance drops, whereas configuration K13 attains the highest mean HV of 0.896 by striking a proper balance between exploration and exploitation. On high-dimensional datasets ($D > 500$), performance disparities widen further due to the combinatorial search space explosion, where configuration K18 secures the best average HV of 0.857 by effectively preserving global search capacity. Consequently, the control parameters of NMACO are adaptively configured according to dataset dimensionality, ensuring optimal search efficiency and convergence across diverse feature space scales.

II. COMPARED LEARNING-BASED FEATURE SELECTION METHODS

To further evaluate the effectiveness of NMACO against learning-based feature selection approaches, we select four representative methods for comparison: MOEA/PSL [1], PTFS

TABLE S2
Average HV performance of the 25 parameter configurations on the low-dimensional datasets.

Parameter	Australian	Ionosphere	SPECT	German	Sonar	Mean
K1	8.06E-1	8.31E-1	6.59E-1	7.91E-1	7.80E-1	7.74E-1
K2	8.09E-1	9.15E-1	7.40E-1	8.34E-1	8.27E-1	8.25E-1
K3	8.12E-1	9.11E-1	7.47E-1	8.33E-1	8.15E-1	8.24E-1
K4	8.16E-1	9.01E-1	7.39E-1	8.32E-1	**8.39E-1**	8.25E-1
K5	8.12E-1	9.00E-1	7.43E-1	8.34E-1	7.95E-1	8.17E-1
K6	8.16E-1	9.00E-1	7.37E-1	8.35E-1	8.17E-1	8.21E-1
K7	8.16E-1	9.14E-1	7.40E-1	8.34E-1	8.27E-1	8.26E-1
K8	8.09E-1	9.04E-1	7.44E-1	8.32E-1	7.84E-1	8.15E-1
K9	8.12E-1	8.94E-1	**7.48E-1**	8.32E-1	8.17E-1	8.21E-1
K10	**8.18E-1**	9.04E-1	7.44E-1	8.33E-1	8.03E-1	8.20E-1
K11	8.07E-1	9.09E-1	7.45E-1	**8.35E-1**	8.26E-1	8.24E-1
K12	8.11E-1	**9.16E-1**	7.44E-1	8.32E-1	8.24E-1	8.25E-1
K13	8.17E-1	9.03E-1	7.44E-1	8.32E-1	8.11E-1	8.21E-1
K14	8.11E-1	9.06E-1	7.44E-1	8.32E-1	7.99E-1	8.19E-1
K15	8.09E-1	8.95E-1	7.44E-1	8.31E-1	8.07E-1	8.17E-1
K16	8.08E-1	9.11E-1	7.41E-1	8.33E-1	8.37E-1	**8.26E-1**
K17	8.11E-1	9.10E-1	7.25E-1	8.34E-1	8.06E-1	8.17E-1
K18	8.13E-1	9.04E-1	7.43E-1	8.33E-1	8.18E-1	8.22E-1
K19	8.15E-1	8.91E-1	7.40E-1	8.32E-1	8.16E-1	8.19E-1
K20	8.12E-1	8.90E-1	7.26E-1	8.32E-1	8.28E-1	8.18E-1
K21	8.07E-1	9.12E-1	7.17E-1	8.35E-1	8.31E-1	8.20E-1
K22	8.15E-1	9.14E-1	7.41E-1	8.33E-1	8.00E-1	8.21E-1
K23	8.12E-1	9.03E-1	7.45E-1	8.32E-1	7.86E-1	8.15E-1
K24	8.11E-1	9.07E-1	7.40E-1	8.31E-1	8.03E-1	8.18E-1
K25	8.11E-1	9.08E-1	7.38E-1	8.31E-1	8.13E-1	8.20E-1

TABLE S3
Average HV performance of the 25 parameter configurations on the medium-dimensional datasets.

Parameter	ULC	Musk	LSVT	Mean
K1	6.88E-1	6.61E-1	6.15E-1	6.55E-1
K2	8.36E-1	8.58E-1	7.88E-1	8.28E-1
K3	8.43E-1	9.02E-1	8.32E-1	8.59E-1
K4	8.48E-1	9.21E-1	8.56E-1	8.75E-1
K5	8.45E-1	9.30E-1	8.56E-1	8.77E-1
K6	7.54E-1	7.17E-1	6.81E-1	7.17E-1
K7	8.41E-1	9.15E-1	8.61E-1	8.72E-1
K8	8.41E-1	9.33E-1	8.66E-1	8.80E-1
K9	8.40E-1	9.42E-1	8.83E-1	8.89E-1
K10	8.46E-1	9.33E-1	8.62E-1	8.80E-1
K11	8.07E-1	7.86E-1	7.19E-1	7.70E-1
K12	**8.55E-1**	9.28E-1	8.96E-1	8.93E-1
K13	8.40E-1	**9.43E-1**	**9.06E-1**	**8.96E-1**
K14	8.52E-1	9.32E-1	8.60E-1	8.82E-1
K15	8.39E-1	9.36E-1	8.96E-1	8.91E-1
K16	8.47E-1	8.44E-1	8.04E-1	8.32E-1
K17	8.50E-1	9.36E-1	8.87E-1	8.91E-1
K18	8.54E-1	9.17E-1	8.43E-1	8.72E-1
K19	8.44E-1	9.38E-1	8.89E-1	8.90E-1
K20	8.35E-1	9.39E-1	8.87E-1	8.87E-1
K21	8.45E-1	8.61E-1	8.39E-1	8.48E-1
K22	8.46E-1	8.86E-1	8.32E-1	8.55E-1
K23	8.46E-1	9.35E-1	8.86E-1	8.89E-1
K24	8.38E-1	9.42E-1	8.97E-1	8.92E-1
K25	8.35E-1	9.39E-1	8.88E-1	8.87E-1

TABLE S4
Average HV performance of the 25 parameter configurations on the high-dimensional datasets.

Parameter	COIL20	Colon	ORL	Yale	SRBCT	Lung	DLBCL	Lymphoma	Leukemia	Mean
K1	5.45E-1	6.25E-1	5.65E-1	6.39E-1	8.24E-1	7.29E-1	7.32E-1	5.93E-1	7.22E-1	6.64E-1
K2	7.66E-1	7.81E-1	7.50E-1	7.10E-1	9.45E-1	9.17E-1	9.15E-1	6.59E-1	8.90E-1	8.15E-1
K3	8.20E-1	8.04E-1	7.83E-1	6.88E-1	9.45E-1	9.34E-1	9.60E-1	7.41E-1	9.18E-1	8.44E-1
K4	8.50E-1	8.14E-1	7.70E-1	6.68E-1	9.61E-1	9.38E-1	9.60E-1	6.85E-1	9.13E-1	8.40E-1
K5	8.61E-1	8.24E-1	7.77E-1	6.53E-1	9.50E-1	9.38E-1	9.65E-1	6.72E-1	9.52E-1	8.44E-1
K6	7.23E-1	7.22E-1	6.36E-1	6.43E-1	8.90E-1	8.67E-1	7.06E-1	6.43E-1	7.94E-1	7.36E-1
K7	8.49E-1	8.13E-1	**8.05E-1**	6.66E-1	9.63E-1	9.40E-1	9.65E-1	6.92E-1	9.35E-1	8.47E-1
K8	8.73E-1	**8.33E-1**	7.81E-1	6.57E-1	9.74E-1	9.43E-1	9.53E-1	6.64E-1	9.49E-1	8.48E-1
K9	8.70E-1	8.22E-1	7.49E-1	5.92E-1	9.40E-1	9.53E-1	9.55E-1	7.11E-1	9.20E-1	8.34E-1
K10	8.67E-1	8.19E-1	7.61E-1	6.51E-1	9.39E-1	9.30E-1	9.53E-1	7.41E-1	9.89E-1	8.50E-1
K11	7.77E-1	7.11E-1	6.80E-1	7.20E-1	9.06E-1	8.82E-1	7.80E-1	5.69E-1	7.85E-1	7.57E-1
K12	8.74E-1	8.20E-1	7.93E-1	6.59E-1	9.44E-1	9.29E-1	9.34E-1	6.69E-1	9.08E-1	8.37E-1
K13	**8.86E-1**	8.25E-1	7.83E-1	6.13E-1	9.61E-1	**9.57E-1**	9.54E-1	6.89E-1	9.19E-1	8.43E-1
K14	8.73E-1	7.98E-1	7.61E-1	6.34E-1	9.47E-1	9.20E-1	9.19E-1	**7.62E-1**	9.63E-1	8.42E-1
K15	8.66E-1	8.05E-1	7.56E-1	6.61E-1	9.10E-1	8.98E-1	9.38E-1	7.64E-1	**9.91E-1**	8.43E-1
K16	7.98E-1	7.30E-1	7.71E-1	7.40E-1	9.21E-1	9.14E-1	8.55E-1	5.65E-1	7.92E-1	7.87E-1
K17	8.76E-1	7.96E-1	7.88E-1	6.20E-1	9.54E-1	9.40E-1	**9.97E-1**	6.87E-1	8.96E-1	8.39E-1
K18	8.67E-1	8.22E-1	7.96E-1	6.58E-1	9.59E-1	9.38E-1	9.94E-1	7.51E-1	9.29E-1	**8.57E-1**
K19	8.76E-1	7.85E-1	7.61E-1	6.68E-1	9.33E-1	9.10E-1	9.38E-1	7.73E-1	9.75E-1	8.46E-1
K20	8.57E-1	7.86E-1	7.77E-1	6.96E-1	9.03E-1	9.10E-1	9.30E-1	7.46E-1	9.88E-1	8.44E-1
K21	8.30E-1	7.10E-1	7.75E-1	**7.45E-1**	9.52E-1	9.24E-1	8.86E-1	5.79E-1	7.99E-1	8.00E-1
K22	8.30E-1	8.08E-1	7.89E-1	7.05E-1	**9.76E-1**	9.44E-1	9.37E-1	6.58E-1	9.28E-1	8.42E-1
K23	8.74E-1	7.85E-1	7.66E-1	6.70E-1	9.32E-1	9.25E-1	9.64E-1	7.45E-1	9.60E-1	8.47E-1
K24	8.73E-1	7.65E-1	7.56E-1	7.04E-1	9.10E-1	9.17E-1	9.04E-1	7.56E-1	9.61E-1	8.38E-1
K25	8.77E-1	7.23E-1	7.63E-1	6.93E-1	8.87E-1	9.10E-1	9.47E-1	7.21E-1	9.63E-1	8.32E-1

[2], QLFS [3] and KLEA [4]. MOEA/PSL is an unsupervised neural network-based multi-objective optimization method, which learns low-dimensional representations through neural networks and searches sparse solutions in the learned latent space. PTFS introduces tensor learning into feature selection and exploits high-order correlations among features to identify informative feature subsets. QLFS incorporates a Q-learning mechanism to model feature selection as a sequential decision-making process, where feature importance is updated through reinforcement learning. KLEA is a knowledge learning-based dimensionality reduction method that transfers useful knowledge from historical search experiences to accelerate the optimization process. These methods cover different learning paradigms for feature selection. All algorithms were independently executed 30 times on the benchmark datasets. The Wilcoxon rank-sum test with Holm-Bonferroni correction was conducted to assess statistical significance.

A. Performance Comparison on HV *and* IGD

Tables S5 and S6 report the HV and IGD results of NMACO and four learning-based feature selection methods. HV evaluates the convergence and diversity of the obtained Pareto solutions, while IGD measures the approximation quality toward the reference Pareto front.

For the HV metric, NMACO achieves the best performance on 14 out of 17 datasets, including Australian, Ionosphere, SPECT, German, Sonar, Urban Land Cover, LSVT, COIL20, Colon, SRBCT, Lung, Lymphoma, DLBCL and Leukemia. Moreover, the Holm-Bonferroni corrected Wilcoxon test demonstrates that NMACO significantly outperforms MOEA/PSL, PTFS, QLFS, and KLEA on 14, 13, 14, and 13 datasets, respectively. Particularly, NMACO obtains clear improvements on high-dimensional datasets. For example, on SRBCT, DLBCL, Lymphoma, and Leukemia, NMACO achieves HV values of 0.965, 0.935, 0.763, and 0.984, respectively, which are considerably higher than those obtained by the learning-based competitors. These results indicate that NMACO can generate Pareto solutions with better convergence and diversity when dealing with complex feature spaces.

The IGD results further verify the effectiveness of NMACO. Compared with the learning-based methods, NMACO obtains lower IGD values on most datasets and achieves significant improvements on challenging datasets such as Ionosphere, Urban Land Cover, SRBCT, DLBCL, Lymphoma, and Leukemia. For instance, on Leukemia, NMACO reduces the IGD value to 0.0148, while MOEA/PSL, PTFS, QLFS, and KLEA obtain 0.0710, 0.0857, 0.0743, and 0.0383, respectively. Similarly, on Lymphoma, NMACO achieves an IGD value of 0.0374, which is substantially smaller than those of MOEA/PSL (0.149), PTFS (0.154), QLFS (0.108), and KLEA (0.0949).

The superior HV and IGD performance can be attributed to the niche-partitioning multi-pheromone updating mechanism. Unlike learning-based approaches that mainly rely on representation learning or knowledge extraction, NMACO explicitly maintains multiple search regions according to feature subset structural similarity. This mechanism reduces the risk of population aggregation toward limited regions and improves the exploration of diverse Pareto-optimal feature subsets.

B. Analysis on Classification Performance and Dimensionality

Tables S7 and S8 present the classification accuracy and the number of selected features obtained by different methods.

Regarding classification accuracy, NMACO achieves the highest accuracy on 9 out of 17 datasets, including Australian, Ionosphere, LSVT, COIL20, ORL, SRBCT, Lung, DLBCL, and Leukemia. The statistical results show that NMACO significantly outperforms MOEA/PSL, PTFS, QLFS, and KLEA on 9, 13, 12, and 8 datasets, respectively. In particular, NMACO demonstrates remarkable advantages on high-dimensional datasets. For example, NMACO achieves accuracies of 96.89%, 94.22%, 85.12%, and 99.21% on SRBCT,

TABLE S5
Average HV results obtained by different methods on the testing sets

Dataset	MOEA/PSL	PTFS	QLFS	KLEA	NMACO
Australian	7.90E-1(1.5E-2)(+)	6.76E-1(1.6E-3)(+)	7.59E-1(3.2E-3)(+)	7.89E-1(1.3E-2)(+)	**8.10E-1(4.0E-3)**
SPECT	7.10E-1(1.9E-2)(+)	6.87E-1(1.8E-3)(+)	6.94E-1(1.6E-2)(+)	7.08E-1(2.7E-2)(+)	**7.35E-1(1.7E-2)**
German	8.28E-1(1.0E-2)(+)	6.07E-1(2.0E-3)(+)	7.99E-1(3.4E-16)(+)	8.22E-1(9.7E-3)(+)	**8.35E-1(4.7E-3)**
Ionosphere	8.65E-1(1.8E-2)(+)	8.23E-1(1.1E-2)(+)	8.30E-1(3.2E-3)(+)	8.64E-1(1.9E-2)(+)	**9.15E-1(8.9E-3)**
Sonar	7.74E-1(3.5E-2)(+)	7.00E-1(3.1E-2)(+)	8.15E-1(1.3E-2)(+)	7.95E-1(3.0E-2)(+)	**8.28E-1(1.9E-2)**
ULC	8.32E-1(2.4E-2)(+)	7.60E-1(7.9E-3)(+)	7.72E-1(2.1E-2)(+)	8.32E-1(1.7E-2)(+)	**8.47E-1(1.5E-2)**
Musk	**9.57E-1(4.2E-3)(-)**	9.43E-1(1.5E-3)(-)	9.53E-1(8.5E-4)(-)	9.57E-1(3.7E-3)(-)	9.39E-1(3.6E-3)
LSVT	7.85E-1(6.9E-2)(+)	8.40E-1(2.2E-2)(+)	7.89E-1(4.3E-2)(+)	8.17E-1(6.1E-2)(+)	**8.94E-1(2.4E-2)**
COIL20	8.35E-1(3.0E-2)(+)	8.50E-1(7.5E-3)(+)	8.41E-1(3.9E-2)(+)	8.44E-1(2.9E-2)(+)	**8.65E-1(8.5E-3)**
ORL	7.90E-1(3.1E-2)(=)	**8.22E-1(1.2E-2)(-)**	7.16E-1(2.4E-2)(+)	7.81E-1(3.1E-2)(+)	7.97E-1(1.7E-2)
Yale	6.31E-1(6.3E-2)(=)	6.28E-1(5.1E-2)(=)	5.42E-1(6.7E-2)(+)	**6.55E-1(6.8E-2)(=)**	6.32E-1(3.4E-2)
Colon	6.95E-1(6.8E-2)(+)	7.17E-1(4.7E-2)(+)	6.33E-1(3.4E-16)(+)	7.09E-1(8.2E-2)(+)	**8.23E-1(6.0E-3)**
SRBCT	7.86E-1(1.1E-1)(+)	8.06E-1(3.0E-2)(+)	8.12E-1(8.5E-2)(+)	7.83E-1(8.2E-2)(+)	**9.65E-1(2.5E-2)**
Lung	9.14E-1(2.6E-2)(+)	9.34E-1(1.8E-2)(=)	9.33E-1(3.4E-16)(=)	9.30E-1(2.0E-2)(=)	**9.36E-1(1.0E-2)**
Lymphoma	6.55E-1(8.0E-2)(+)	6.92E-1(5.2E-2)(+)	7.60E-1(0.0E+0)(=)	7.42E-1(5.8E-2)(=)	**7.63E-1(2.3E-2)**
DLBCL	8.42E-1(6.5E-2)(+)	7.18E-1(2.7E-2)(+)	7.38E-1(1.1E-16)(+)	8.63E-1(7.1E-2)(+)	**9.35E-1(2.5E-2)**
Leukemia	8.60E-1(5.6E-2)(+)	8.66E-1(2.2E-2)(+)	9.01E-1(2.3E-16)(+)	9.02E-1(6.0E-2)(+)	**9.84E-1(2.1E-2)**
+/-/=	14/1/2	13/2/2	14/1/2	13/1/3	—
Rank	3.35	3.71	3.65	2.94	1.35

TABLE S6
Average IGD results obtained by different methods on the testing sets

Dataset	MOEA/PSL	PTFS	QLFS	KLEA	NMACO
Australian	1.77E-2(9.6E-3)(+)	2.28E-1(0.0E+0)(+)	8.07E-2(5.8E-3)(+)	1.49E-2(9.7E-3)(=)	**1.21E-2(7.4E-3)**
SPECT	2.76E-2(9.0E-3)(=)	9.14E-2(2.8E-3)(+)	7.10E-2(1.0E-2)(+)	2.62E-2(9.0E-3)(=)	**2.47E-2(8.8E-3)**
German	1.38E-2(6.7E-3)(-)	2.57E-1(3.5E-4)(+)	3.10E-2(1.1E-17)(=)	**1.10E-2(5.4E-3)(-)**	2.78E-2(1.7E-2)
Ionosphere	3.77E-2(1.1E-2)(+)	9.02E-2(8.6E-3)(+)	5.63E-2(9.2E-3)(+)	3.96E-2(1.3E-2)(+)	**1.32E-2(8.7E-3)**
Sonar	3.59E-2(1.1E-2)(+)	9.84E-2(2.0E-2)(+)	6.23E-2(1.5E-2)(+)	3.64E-2(1.5E-2)(+)	**2.82E-2(9.6E-3)**
ULC	4.46E-2(1.5E-2)(+)	6.38E-2(1.2E-2)(+)	7.93E-2(1.9E-2)(+)	4.20E-2(1.0E-2)(+)	**1.66E-2(9.0E-3)**
Musk	3.16E-2(5.8E-3)(-)	**2.50E-2(5.5E-3)(-)**	2.84E-2(3.8E-3)(-)	3.33E-2(2.0E-3)(-)	4.81E-2(1.7E-2)
LSVT	2.76E-2(2.4E-2)(=)	7.80E-2(3.2E-2)(+)	5.41E-2(1.7E-2)(+)	2.30E-2(1.2E-2)(=)	**1.88E-2(1.1E-2)**
COIL20	4.07E-2(1.6E-2)(+)	4.15E-2(1.2E-2)(+)	4.93E-2(2.3E-2)(+)	3.39E-2(1.5E-2)(+)	**2.22E-2(7.4E-3)**
ORL	**1.94E-2(1.6E-2)(-)**	2.42E-2(4.0E-3)(-)	6.73E-2(1.6E-2)(+)	2.12E-2(6.8E-3)(-)	4.93E-2(1.2E-2)
Yale	3.06E-2(2.8E-2)(=)	3.15E-2(9.2E-3)(=)	1.01E-1(4.1E-2)(+)	**2.01E-2(1.0E-2)(-)**	3.07E-2(9.1E-3)
Colon	5.77E-2(3.1E-2)(=)	5.38E-2(2.4E-2)(+)	**2.73E-2(0.0E+0)(-)**	3.92E-2(1.8E-2)(=)	3.91E-2(1.2E-2)
SRBCT	8.86E-2(4.7E-2)(+)	9.99E-2(1.0E-2)(+)	7.94E-2(2.8E-2)(+)	8.28E-2(3.6E-2)(+)	**6.04E-2(2.0E-2)**
Lung	2.12E-2(1.5E-2)(=)	1.78E-2(7.4E-3)(=)	2.51E-2(7.1E-18)(+)	**1.33E-2(5.8E-3)(=)**	1.49E-2(6.1E-3)
Lymphoma	1.49E-1(4.9E-2)(+)	1.54E-1(4.5E-2)(+)	1.08E-1(2.8E-17)(+)	9.49E-2(2.7E-2)(+)	**3.74E-2(9.9E-3)**
DLBCL	3.04E-2(2.2E-2)(=)	1.14E-1(3.1E-2)(+)	1.30E-1(0.0E+0)(+)	5.29E-2(2.9E-2)(+)	**2.24E-2(1.3E-2)**
Leukemia	7.10E-2(4.2E-2)(+)	8.57E-2(2.0E-2)(+)	7.43E-2(4.2E-17)(+)	3.83E-2(3.3E-2)(+)	**1.48E-2(1.2E-2)**
+/-/=	8/3/6	13/2/2	14/2/1	8/4/5	—
Rank	2.88	4.24	3.88	2.24	1.76

Lung, Lymphoma, and Leukemia, respectively. Compared with existing learning-based methods, the improvements are especially significant on SRBCT and Leukemia, where NMACO provides more than 10% and 4% accuracy gains over most competitors.

Although some learning-based methods obtain slightly fewer selected features on several datasets, NMACO provides a better balance between feature reduction and classification performance. Specifically, NMACO achieves lower feature numbers than PTFS, QLFS, and KLEA on most datasets, while maintaining higher classification accuracy. For example, on Australian and German, NMACO selects only 4.23 and 4.17 features on average while achieving accuracies of 89.41% and 87.44%, respectively. On Leukemia, NMACO selects 69.5 features, which is considerably fewer than PTFS (820 features) and QLFS (156 features), while obtaining the highest accuracy of 99.21%.

The effectiveness of NMACO in balancing sparsity and predictive performance benefits from the heuristic guidance strategy and subset repair mechanism. The heuristic guidance strategy introduces feature-label relevance and feature-feature redundancy information into the transition probability, helping

TABLE S7
Average ACC% results obtained by different methods on the testing sets

Dataset	MOEA/PSL	PTFS	QLFS	KLEA	NMACO
Australian	88.71%(1.22)(+)	87.02%(0.00)(+)	87.02%(0.00)(+)	88.65%(0.88)(+)	**89.40%(0.58)**
SPECT	79.21%(2.56)(=)	76.89%(0.57)(+)	**79.41%(2.04)(=)**	79.21%(1.66)(=)	79.31%(2.15)
German	**87.77%(0.89)(=)**	82.33%(0.00)(+)	85.67%(0.00)(+)	87.72%(0.71)(=)	87.45%(0.60)
Ionosphere	91.66%(2.20)(+)	85.41%(1.31)(+)	88.38%(0.95)(+)	91.24%(2.32)(+)	**94.59%(0.94)**
Sonar	82.88%(2.57)(+)	80.75%(7.83)(+)	**87.19%(2.02)(-)**	84.56%(3.26)(=)	84.88%(2.15)
ULC	**85.77%(1.33)(=)**	81.55%(0.00)(+)	80.10%(1.65)(+)	**85.77%(1.43)(=)**	85.57%(1.51)
Musk	**96.85%(0.34)(-)**	96.53%(0.22)(-)	96.76%(0.13)(-)	96.78%(0.37)(-)	95.72%(0.28)
LSVT	85.95%(4.63)(+)	87.49%(2.14)(+)	82.87%(3.29)(+)	87.69%(3.14)(+)	**90.15%(2.53)**
COIL20	85.20%(2.73)(+)	86.60%(0.80)(+)	85.72%(4.06)(+)	86.08%(2.53)(+)	**90.85%(0.70)**
ORL	81.17%(2.69)(=)	**84.83%(0.93)(-)**	78.77%(1.02)(+)	80.17%(2.73)(=)	80.73%(1.82)
Yale	66.27%(6.07)(=)	64.80%(4.06)(=)	61.07%(7.87)(=)	**68.53%(6.02)(-)**	63.40%(3.45)
Colon	75.37%(5.63)(+)	77.68%(5.54)(+)	78.95%(0.00)(+)	77.05%(3.68)(+)	**84.21%(0.00)**
SRBCT	84.30%(7.35)(+)	84.00%(2.77)(+)	85.63%(5.49)(+)	85.78%(5.82)(+)	**96.89%(2.55)**
Lung	92.76%(2.64)(=)	93.84%(1.85)(=)	93.65%(0.00)(=)	93.90%(1.98)(=)	**94.22%(1.01)**
Lymphoma	71.25%(6.25)(+)	70.00%(4.94)(+)	78.12%(0.00)(+)	78.25%(4.08)(+)	**85.12%(2.26)**
DLBCL	91.65%(5.01)(=)	76.35%(4.18)(+)	73.91%(0.00)(+)	**93.91%(5.61)(=)**	93.57%(2.55)
Leukemia	91.62%(5.19)(+)	91.24%(1.78)(+)	90.48%(0.00)(+)	95.24%(0.00)(+)	**99.24%(1.78)**
+/-/=	9/1/7	13/2/2	12/2/3	8/2/7	—
Rank	3.06	3.85	3.74	2.41	1.94

ants identify informative features. Meanwhile, the subset repair mechanism further removes redundant feature combinations during local refinement, improving the compactness of the final selected subsets.

Overall, the additional experiments against learning-based feature selection methods demonstrate that NMACO not only achieves competitive optimization performance but also provides more discriminative and compact feature subsets for classification tasks.

III. Compared With NMDE and NMDE-JN

To further investigate the relationship between the proposed diversity preservation strategy and the underlying optimization framework, additional comparisons among NMACO, NMDE [5], and NMDE-JN are conducted on all 17 datasets. NMDE is selected as a representative niching-based multi-objective feature selection algorithm, while NMDE-JN is constructed by replacing the original niching mechanism of NMDE with the dynamic niching partition strategy proposed in this work.

A. Performance Comparison on HV and IGD

Tables S9 and S10 report the HV and IGD results of NMACO and NMDE, NMDE-JN and NMACO methods. HV evaluates the convergence and diversity of the obtained Pareto solutions, while IGD measures the approximation quality toward the reference Pareto front.

The HV results are presented in Table S9. NMACO achieves the best average rank on HV, ranking first among the three methods. NMDE ranks second, and NMDE-JN ranks third. This ranking indicates that the niching strategy alone does not improve HV performance when it is decoupled from the ACO framework. The highest HV values on all 17 datasets are obtained by NMACO, and both NMDE and NMDE-JN are significantly outperformed according to the Holm-Bonferroni corrected statistical test. Improvements over NMDE are observed to range from approximately 6.7% (Yale) to 72.0% (DLBCL). These results indicate that the quality of the obtained Pareto solutions is effectively enhanced by NMACO in terms of both convergence and diversity. Consistent improvements over NMDE-JN are also achieved on all datasets.

The comparison between NMDE and NMDE-JN reveals an important observation. A worse average rank is obtained by NMDE-JN than by standard NMDE, which suggests that diversity preservation alone is insufficient to enhance solution quality. The additional improvements achieved by NMACO over both NMDE and NMDE-JN indicate that the performance gains of NMACO are derived from the combination between niche-guided diversity preservation and the multi-pheromone ACO search framework, rather than from the niching mechanism in isolation.

The IGD results are presented in Table S10, and a consistent conclusion is provided. The best average rank on IGD is achieved by NMACO, which ranks first among the three methods. NMDE ranks second, and NMDE-JN ranks third. The lowest IGD values on all 17 datasets are obtained by NMACO, and both NMDE and NMDE-JN are significantly outperformed. For example, on the high-dimensional datasets DLBCL, Lung, and Leukemia, IGD values of 0.0224, 0.0149, and 0.0148 are obtained by NMACO, respectively, which are substantially lower than those obtained by NMDE and NMDE-JN. These results demonstrate that the Pareto-optimal region can be approached more closely by NMACO while a better solution distribution is maintained. The consistent improve-

TABLE S8
Average FN results obtained by different methods on the testing sets

Dataset	MOEA/PSL	PTFS	QLFS	KLEA	NMACO
Australian	4.32E+0(6.9E-1)(=)	7.00E+0(0.0E+0)(+)	9.20E+0(2.4E+0)(+)	4.72E+0(9.4E-1)(=)	**4.28E+0(5.4E-1)**
SPECT	5.68E+0(1.4E+0)(=)	8.72E+0(4.6E-1)(+)	1.01E+1(3.1E+0)(+)	5.44E+0(1.0E+0)(=)	**5.00E+0(5.0E-1)**
German	5.56E+0(1.1E+0)(+)	2.10E+1(0.0E+0)(+)	8.00E+0(0.0E+0)(+)	5.48E+0(1.4E+0)(+)	**4.16E+0(1.4E+0)**
Ionosphere	2.92E+0(7.0E-1)(=)	5.28E+0(3.3E+0)(+)	3.32E+0(1.1E+0)(=)	2.80E+0(6.5E-1)(=)	**2.60E+0(5.8E-1)**
Sonar	**5.28E+0(2.4E+0)(=)**	2.68E+1(2.4E+1)(+)	2.13E+1(1.2E+1)(+)	7.88E+0(3.8E+0)(+)	5.92E+0(2.2E+0)
ULC	7.24E+0(2.5E+0)(+)	**3.00E+0(0.0E+0)(-)**	2.62E+1(2.0E+1)(+)	7.40E+0(3.0E+0)(+)	4.16E+0(1.3E+0)
Musk	2.64E+1(8.3E+0)(+)	1.38E+2(2.4E+1)(+)	1.18E+2(2.6E+1)(+)	2.60E+1(9.1E+0)(+)	**2.02E+1(7.1E+0)**
LSVT	**4.08E+0(2.4E+0)(-)**	5.71E+1(2.3E+1)(+)	8.92E+0(1.2E+1)(=)	6.52E+0(8.5E+0)(=)	8.12E+0(3.4E+0)
COIL20	**3.06E+1(1.6E+1)(-)**	1.71E+2(6.5E+1)(+)	1.60E+2(1.6E+2)(+)	4.96E+1(3.8E+1)(-)	8.47E+1(1.5E+1)
ORL	**3.13E+1(2.2E+1)(=)**	1.50E+2(3.3E+1)(+)	7.97E+2(2.8E+2)(+)	5.96E+1(4.1E+1)(+)	3.58E+1(7.1E+0)
Yale	**1.26E+1(8.6E+0)(=)**	3.89E+1(1.1E+1)(+)	3.22E+2(2.3E+2)(+)	2.92E+1(2.7E+1)(=)	1.53E+1(1.1E+1)
Colon	8.00E+0(1.4E+1)(-)	5.63E+1(4.7E+1)(-)	1.80E+1(0.0E+0)(-)	**4.80E+0(6.5E+0)(-)**	1.37E+2(6.5E+1)
SRBCT	1.77E+1(4.3E+1)(-)	2.09E+2(1.7E+2)(+)	1.84E+1(3.6E+1)(-)	**1.69E+1(3.4E+1)(-)**	2.02E+1(8.3E+0)
Lung	**3.29E+1(3.4E+1)(=)**	1.23E+2(1.2E+2)(+)	2.64E+2(0.0E+0)(+)	1.15E+2(1.9E+2)(=)	3.64E+1(9.0E+0)
Lymphoma	4.11E+1(1.0E+2)(-)	**3.94E+1(3.5E+1)(-)**	7.79E+2(0.0E+0)(+)	4.12E+1(2.9E+1)(-)	4.27E+2(4.1E+1)
DLBCL	2.22E+1(7.9E+1)(+)	4.49E+1(1.4E+2)(+)	**1.00E+0(0.0E+0)(-)**	1.11E+1(5.0E+1)(-)	1.26E+1(7.9E+0)
Leukemia	6.72E+1(1.6E+2)(=)	8.35E+2(1.6E+3)(+)	1.56E+2(0.0E+0)(+)	1.37E+2(3.4E+2)(=)	**6.63E+1(2.9E+1)**
+/-/=	4/5/8	14/3/0	12/3/2	5/5/7	—
Rank	2.06	4.18	4.12	2.41	2.24

ments in both HV and IGD indicate that the performance gains of NMACO are not caused by optimizing a single objective, but rather by a better balance between feature reduction and classification performance.

Overall, further insights into the contributions of the diversity preservation mechanism and the search framework are provided by these comparisons. It is shown that the niching strategy alone (NMDE-JN) does not consistently improve HV over the standard DE baseline (NMDE), while the best performance across all datasets is achieved by NMACO. This confirms that the performance gains of NMACO stem from the integration of niche-guided diversity preservation with the multi-pheromone ACO framework and heuristic guidance mechanism, rather than from any single component in isolation.

B. Analysis on Classification Performance and Dimensionality

To further analyze the differences among NMDE, NMDE-JN, and NMACO from the perspectives of classification performance and feature reduction, additional comparisons on ACC and FN are conducted. Since NMDE-JN only replaces the original niching mechanism of NMDE while retaining the DE-based search operator, these results can further reveal the influence of the proposed niching partition strategy and the ACO-based search framework.

As shown in Tables S11 and S12, NMACO achieves the best overall performance in both classification accuracy and feature reduction. For ACC, NMACO ranks first with an average rank of 1.76, followed by NMDE-JN in second and NMDE in third, and achieves the highest accuracy on most datasets. Although NMDE-JN improves ACC over NMDE on several datasets, indicating that the proposed niching partition strategy can help maintain diverse and discriminative feature subsets, its advantages are not consistent across all datasets.

TABLE S9
Average HV results obtained by different methods on the testing sets

Dataset	NMDE	NMDE-JN	NMACO
Australian	6.08E-1(2.6E-2)(+)	5.12E-1(5.8E-2)(+)	**8.10E-1(4.0E-3)**
SPECT	5.87E-1(1.9E-2)(+)	5.40E-1(4.4E-2)(+)	**7.35E-1(1.7E-2)**
German	5.88E-1(6.1E-2)(+)	5.50E-1(7.7E-2)(+)	**8.35E-1(4.7E-3)**
Ionosphere	8.01E-1(2.9E-2)(+)	7.78E-1(3.0E-2)(+)	**9.15E-1(8.9E-3)**
Sonar	5.49E-1(5.5E-2)(+)	7.12E-1(3.5E-2)(+)	**8.28E-1(1.9E-2)**
ULC	2.64E-1(1.2E-1)(+)	5.97E-1(1.8E-2)(+)	**8.47E-1(1.5E-2)**
Musk	9.03E-1(1.1E-2)(+)	7.41E-1(1.9E-2)(+)	**9.39E-1(3.6E-3)**
LSVT	7.50E-1(6.2E-2)(+)	5.96E-1(3.0E-2)(+)	**8.94E-1(2.4E-2)**
COIL20	7.15E-1(1.2E-2)(+)	5.65E-1(8.2E-3)(+)	**8.65E-1(8.5E-3)**
ORL	6.36E-1(2.8E-2)(+)	5.17E-1(1.2E-2)(+)	**7.97E-1(1.7E-2)**
Yale	5.56E-1(3.4E-2)(+)	4.52E-1(2.5E-2)(+)	**6.32E-1(3.4E-2)**
Colon	5.41E-1(4.0E-2)(+)	4.59E-1(3.5E-2)(+)	**8.23E-1(6.0E-3)**
SRBCT	6.72E-1(4.0E-2)(+)	5.75E-1(2.3E-2)(+)	**9.65E-1(2.5E-2)**
Lung	6.98E-1(1.2E-2)(+)	6.08E-1(7.8E-3)(+)	**9.36E-1(1.0E-2)**
Lymphoma	5.78E-1(1.2E-2)(+)	5.28E-1(1.5E-2)(+)	**7.63E-1(2.3E-2)**
DLBCL	6.13E-1(3.2E-2)(+)	5.39E-1(1.6E-2)(+)	**9.35E-1(2.5E-2)**
Leukemia	5.89E-1(8.3E-2)(+)	5.18E-1(7.8E-2)(+)	**9.84E-1(2.1E-2)**
+/-/=	17/0/0	17/0/0	-
Rank	2.12	2.88	1.00

In contrast, NMACO further improves the classification performance in most cases, demonstrating that the combination of pheromone-based cooperative search and heuristic guidance can better exploit informative feature combinations. For FN, NMACO ranks first with an average rank of 1.24, significantly outperforming NMDE and NMDE-JN, and obtains more compact feature subsets on most datasets. Compared with NMDE-JN, the lower FN values of NMACO indicate that the proposed ACO framework can effectively eliminate redundant features while maintaining competitive classification performance. These results suggest that the performance improvement of NMACO is not merely caused by the diversity

TABLE S10
Average IGD results obtained by different methods on the testing sets

Dataset	NMDE	NMDE-JN	NMACO
Australian	1.84E-1(1.4E-2)(+)	1.71E-1(5.0E-2)(+)	**1.21E-2(7.4E-3)**
SPECT	2.57E-1(3.7E-2)(+)	1.65E-1(6.8E-2)(+)	**2.78E-2(1.7E-2)**
German	8.67E-2(1.4E-2)(+)	8.36E-2(2.3E-2)(+)	**1.32E-2(8.7E-3)**
Ionosphere	1.21E-1(5.0E-3)(+)	1.10E-1(3.1E-2)(+)	**2.47E-2(8.8E-3)**
Sonar	1.29E-1(2.4E-2)(+)	1.18E-1(2.0E-2)(+)	**2.82E-2(9.6E-3)**
ULC	1.94E-1(4.2E-2)(+)	2.50E-1(1.4E-2)(+)	**1.66E-2(9.0E-3)**
Musk	7.89E-2(9.6E-3)(+)	1.93E-1(2.0E-2)(+)	**4.81E-2(1.7E-2)**
LSVT	4.01E-2(7.7E-3)(+)	2.92E-1(1.4E-2)(+)	**1.88E-2(1.1E-2)**
COIL20	1.26E-1(5.1E-3)(+)	2.98E-1(7.9E-3)(+)	**2.22E-2(7.4E-3)**
ORL	1.99E-1(6.3E-3)(+)	3.70E-1(8.1E-3)(+)	**4.93E-2(1.2E-2)**
Yale	2.08E-1(8.7E-3)(+)	3.82E-1(1.3E-2)(+)	**3.07E-2(9.1E-3)**
Colon	2.53E-1(7.5E-3)(+)	3.56E-1(8.7E-3)(+)	**3.91E-2(1.2E-2)**
SRBCT	2.75E-1(6.8E-3)(+)	4.03E-1(6.0E-3)(+)	**6.04E-2(2.0E-2)**
Lung	2.70E-1(4.8E-3)(+)	3.69E-1(4.4E-3)(+)	**1.49E-2(6.1E-3)**
Lymphoma	2.22E-1(5.2E-3)(+)	2.92E-1(6.5E-3)(+)	**3.74E-2(9.9E-3)**
DLBCL	2.97E-1(3.9E-3)(+)	3.83E-1(6.2E-3)(+)	**2.24E-2(1.3E-2)**
Leukemia	3.23E-1(2.6E-2)(+)	3.86E-1(2.1E-2)(+)	**1.48E-2(1.2E-2)**
+/-/=	17/0/0	17/0/0	-
Rank	2.29	2.71	1.00

TABLE S11
Average ACC results obtained by different methods on the testing sets

Dataset	NMDE	NMDE-JN	NMACO
Australian	67.90%(2.83)(+)	85.12%(6.39)(+)	**89.40%(0.58)**
SPECT	63.60%(1.71)(+)	72.84%(3.09)(+)	**79.31%(2.15)**
German	64.43%(8.33)(+)	81.28%(7.73)(+)	**87.45%(0.60)**
Ionosphere	85.52%(2.44)(+)	89.52%(1.20)(+)	**94.59%(0.94)**
Sonar	70.31%(2.55)(+)	**90.81%(1.77)(-)**	84.88%(2.15)
ULC	56.99%(5.43)(+)	81.24%(1.22)(+)	**85.57%(1.51)**
Musk	93.66%(0.81)(+)	**96.53%(0.27)(-)**	95.72%(0.28)
LSVT	89.44%(2.80)(=)	**91.90%(1.60)(-)**	90.15%(2.53)
COIL20	88.55%(0.93)(+)	87.76%(0.53)(+)	**90.85%(0.70)**
ORL	81.63%(1.47)(=)	**83.03%(0.93)(-)**	80.73%(1.82)
Yale	**77.67%(2.26)(-)**	76.27%(2.37)(-)	63.40%(3.45)
Colon	**84.63%(2.60)(=)**	83.37%(2.91)(=)	84.21%(0.00)
SRBCT	**97.19%(2.21)(=)**	96.74%(2.22)(=)	96.89%(2.55)
Lung	**98.22%(0.53)(-)**	98.10%(0.65)(-)	94.22%(1.01)
Lymphoma	86.75%(1.36)(-)	**87.25%(0.87)(-)**	85.12%(2.26)
DLBCL	92.52%(1.99)(=)	91.30%(1.77)(+)	**93.57%(2.55)**
Leukemia	94.10%(5.71)(+)	94.10%(5.71)(+)	**99.24%(1.78)**
+/-/=	9/3/5	8/7/2	-
Rank	2.24	2.00	1.76

preservation strategy, but also benefits from the proposed multi-pheromone search mechanism and heuristic guidance.

Overall, the ACC and FN results demonstrate that the superiority of NMACO is not achieved by sacrificing feature reduction for classification performance. Instead, NMACO effectively balances classification accuracy and subset compactness. The comparisons among NMDE, NMDE-JN, and NMACO further verify that the proposed habitat partitioning strategy contributes to diversity maintenance, while the ACO-based search framework provides additional capability for identifying accurate and compact feature subsets.

TABLE S12
Average FN results obtained by different methods on the testing sets

Dataset	NMDE	NMDE-JN	NMACO
Australian	**4.04E+0(1.5E+0)(=)**	5.60E+0(6.5E-1)(+)	4.28E+0(5.4E-1)
SPECT	5.52E+0(1.4E+0)(=)	7.32E+0(2.0E+0)(+)	**5.00E+0(5.0E-1)**
German	5.92E+0(2.5E+0)(+)	8.68E+0(1.4E+0)(+)	**4.16E+0(1.4E+0)**
Ionosphere	3.36E+0(1.2E+0)(+)	6.64E+0(2.4E+0)(+)	**2.60E+0(5.8E-1)**
Sonar	**1.00E+0(0.0E+0)(-)**	2.11E+1(4.4E+0)(+)	5.92E+0(2.2E+0)
ULC	**1.04E+0(2.0E-1)(-)**	5.31E+1(6.4E+0)(+)	4.16E+0(1.3E+0)
Musk	**6.44E+0(1.4E+0)(-)**	5.80E+1(6.0E+0)(+)	2.02E+1(7.1E+0)
LSVT	1.47E+1(2.6E+0)(+)	1.15E+2(1.2E+1)(+)	**8.12E+0(3.4E+0)**
COIL20	1.85E+2(7.7E+0)(+)	4.02E+2(1.7E+1)(+)	**8.47E+1(1.5E+1)**
ORL	1.85E+2(7.5E+0)(+)	4.12E+2(2.1E+1)(+)	**3.58E+1(7.1E+0)**
Yale	1.86E+2(6.5E+0)(+)	4.10E+2(1.8E+1)(+)	**1.53E+1(1.1E+1)**
Colon	5.36E+2(1.8E+1)(+)	7.69E+2(2.7E+1)(+)	**1.37E+2(6.5E+1)**
SRBCT	5.76E+2(1.6E+1)(+)	9.20E+2(4.0E+1)(+)	**2.02E+1(8.3E+0)**
Lung	9.08E+2(1.8E+1)(+)	1.33E+3(5.0E+1)(+)	**3.64E+1(9.0E+0)**
Lymphoma	1.24E+3(2.7E+1)(+)	1.59E+3(5.1E+1)(+)	**4.27E+2(4.1E+1)**
DLBCL	1.63E+3(2.5E+1)(+)	2.21E+3(6.6E+1)(+)	**1.26E+1(7.9E+0)**
Leukemia	2.26E+3(7.0E+1)(+)	2.71E+3(5.5E+1)(+)	**6.63E+1(2.9E+1)**
+/-/=	12/3/2	17/0/0	-
Rank	1.76	3.00	1.24

TABLE S13
Average HV and ACC Performance of NMACO Variant

No.	HV		ACC (%)	
	wo-MP	NMACO	wo-MP	NMACO
Australian	7.82E-1(+)	**8.10E-1**	89.19(=)	**89.40**
SPECT	6.07E-1(+)	**7.35E-1**	**79.41(=)**	79.31
German	8.23E-1(+)	**8.35E-1**	**87.56(=)**	87.45
Ionosphere	8.54E-1(+)	**9.15E-1**	90.70(+)	**94.59**
Sonar	7.82E-1(+)	**8.28E-1**	83.00(+)	**84.88**
ULC	7.93E-1(+)	**8.47E-1**	**85.63(=)**	85.57
Musk	8.23E-1(+)	**9.39E-1**	**96.80(-)**	95.72
LSVT	7.94E-1(+)	**8.94E-1**	85.03(+)	**90.15**
COIL20	7.17E-1(+)	**8.65E-1**	89.23(+)	**90.85**
ORL	7.58E-1(+)	**7.97E-1**	**83.17(-)**	80.73
Yale	**6.94E-1(-)**	6.32E-1	**73.47(-)**	63.40
Colon	6.92E-1(+)	**8.23E-1**	77.89(+)	**84.21**
SRBCT	8.13E-1(+)	**9.65E-1**	83.56(+)	**96.89**
Lung	9.25E-1(+)	**9.36E-1**	**95.62(-)**	94.22
Lymphoma	4.31E-1(+)	**7.63E-1**	82.75(+)	**85.12**
DLBCL	**9.50E-1(=)**	9.35E-1	**96.70(-)**	93.57
Leukemia	7.94E-1(+)	**9.84E-1**	94.86(+)	**99.24**
+/-/=	15/1/1	—	8/5/4	—
Rank	1.88	1.12	1.53	1.47

IV. Ablation Study of NMACO-wo-MP

The niche partitioning strategy is retained, while the multiple pheromone matrices are replaced with a single global pheromone matrix shared by all ants. The method is termed NMACO-wo-MP.

In Table S13, NMACO outperforming NMACO-wo-MP on 15 out of 17 datasets according to the Wilcoxon test in terms of the HV metric and securing average Friedman ranks first for HV and ACC, respectively. As illustrated in Fig. 1, the single pheromone matrix in NMACO-wo-MP suffers from conflicting objective interference, leading to degenerated solution distributions. In contrast, the multiple pheromone matrix update mechanism enables NMACO to successfully discover alternative feature subsets, as evidenced by the well-distributed

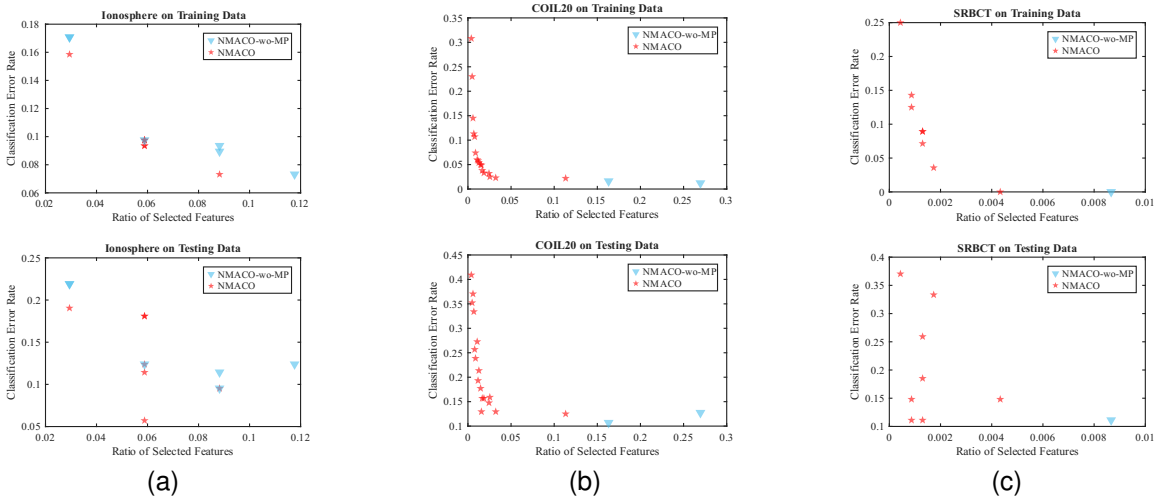


Fig. 1: Evolved PFs by NMACO-wo-MP and NMACO on the Ionosphere, COIL20, and SRBCT datasets.

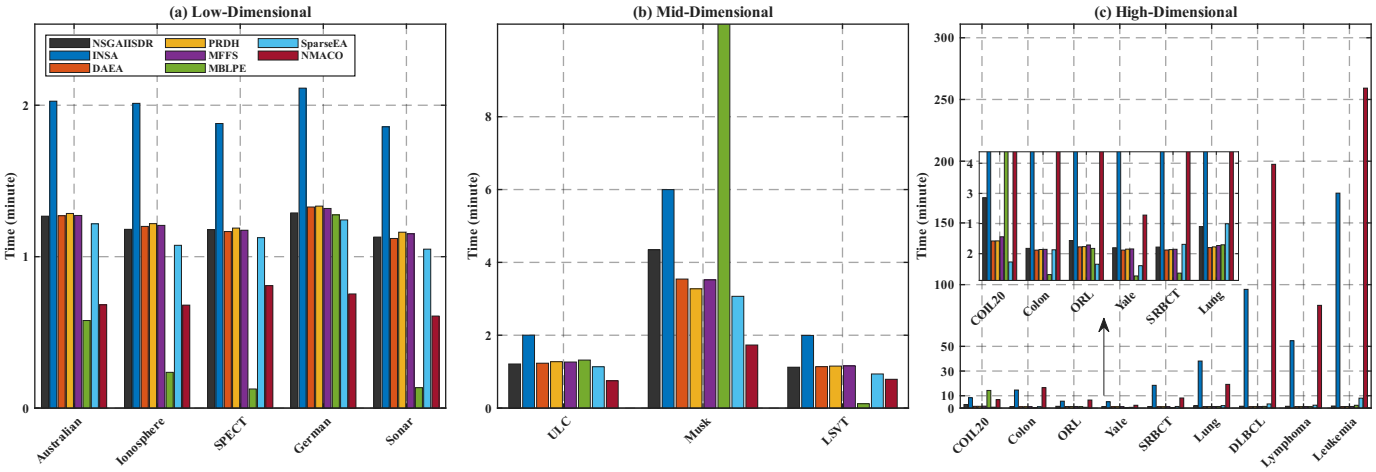


Fig. 2: Wall-clock execution time comparisons (in minutes) across Low-D, Mid-D, and High-D datasets: (a) Low-Dimensional, (b) Mid-Dimensional, and (c) High-Dimensional.

non-dominated solutions on Ionosphere, COIL20, and SRBCT.

V. Runtime Comparison and Computational Analysis

To further evaluate the practical computational efficiency of NMACO, the average wall-clock execution time over 30 independent runs is compared with other algorithms on datasets with different feature dimensionalities, as shown in Fig. 2. The datasets are divided into low-, mid-, and high-dimensional groups to provide an intuitive comparison of the computational characteristics of different methods.

From Fig. 2, NMACO achieves competitive runtime performance compared with most baseline algorithms on low- and medium-dimensional datasets. For example, on the Urban Land Cover dataset, NMACO completes the optimization within 1.8 minutes, whereas MBLPE requires over 533 minutes. This indicates that the additional operations introduced by the multi-pheromone mechanism and niche management do not impose significant computational burden when the feature dimension is moderate.

For high-dimensional datasets, NMACO requires more computational time. This overhead mainly results from the maintenance of multiple pheromone matrices and the increased number of classifier evaluations in a large search space. Nevertheless, this additional computational cost is justified by the improved quality of the obtained Pareto solution sets, as demonstrated by the consistent improvements in HV, IGD, ACC, and FN. Therefore, a reasonable trade-off between computational cost and optimization performance is achieved by NMACO for high-dimensional multi-objective feature selection.

VI. Additional Pareto Front Comparisons

In addition to the four representative datasets shown in Fig. 6 of the main text, the median Pareto fronts obtained by all compared algorithms on the remaining datasets are provided in Fig. S1–Fig. S3. These results further confirm the observations discussed in the main text.

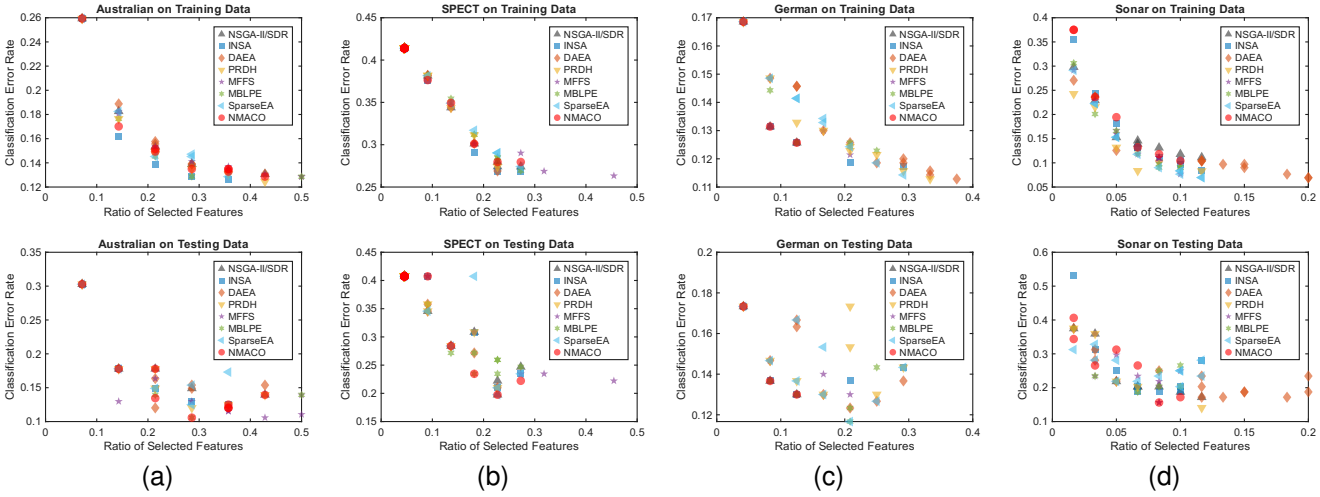


Fig. 3: Median Pareto fronts obtained by all compared algorithms on the remaining datasets (Part 1): (a) Australian, (b) SPECT, (c) German, and (d) Sonar.

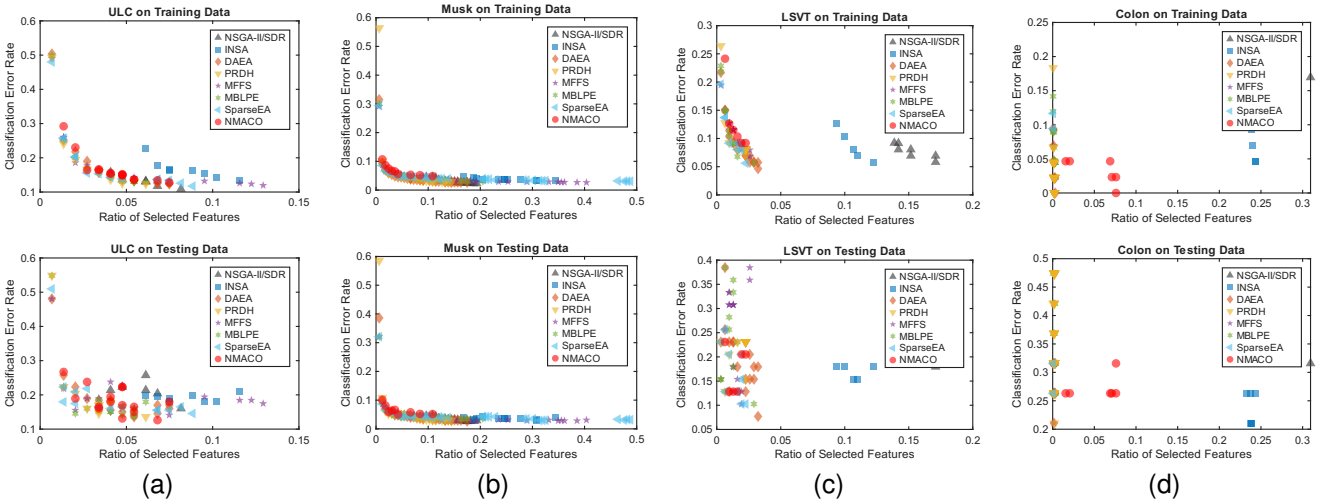


Fig. 4: Median Pareto fronts obtained by all compared algorithms on the remaining datasets (Part 2): (a) ULC, (b) Musk, (c) LSVT, and (d) Colon.

REFERENCES

[1] Y. Tian, C. Lu, X. Zhang, K. C. Tan, and Y. Jin, "Solving large-scale multiobjective optimization problems with sparse optimal solutions via unsupervised neural networks," *IEEE Transactions on Cybernetics*, vol. 51, no. 6, pp. 3115–3128, 2021.

[2] Z. Cao and X. Xie, "Partition-level tensor learning-based multiview unsupervised feature selection," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 36, no. 7, pp. 12 799–12 811, 2025.

[3] R. Xu, M. Li, Z. Yang *et al.*, "Dynamic feature selection algorithm based on q-learning mechanism," *Applied Intelligence*, vol. 51, no. 10, pp. 7233–7244, 2021.

[4] S. Shao, Y. Tian, Y. Zhang, and X. Zhang, "Knowledge learning-based dimensionality reduction for solving large-scale sparse multiobjective optimization problems," *IEEE Transactions on Cybernetics*, vol. 55, no. 7, pp. 3471–3484, 2025.

[5] P. Wang, B. Xue, J. Liang, and M. Zhang, "Differential evolution-based feature selection: A niching-based multiobjective approach," *IEEE Transactions on Evolutionary Computation*, vol. 27, no. 2, pp. 296–310, 2023.

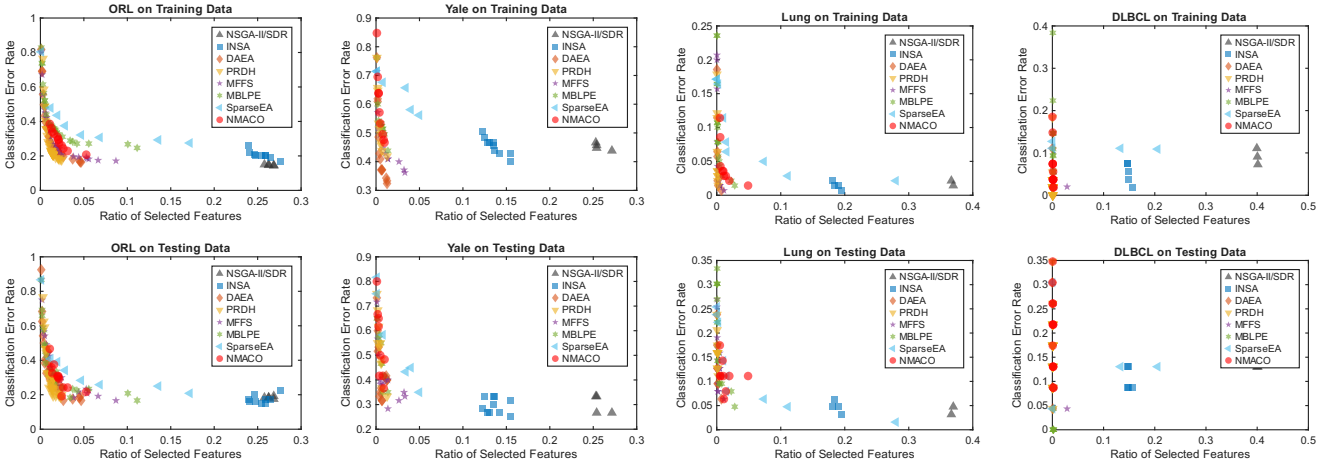


Fig. 5: Median Pareto fronts obtained by all compared algorithms on the remaining datasets (Part 3): (a) ORL, (b) Yale, (c) Lung, and (d) DLBCL.

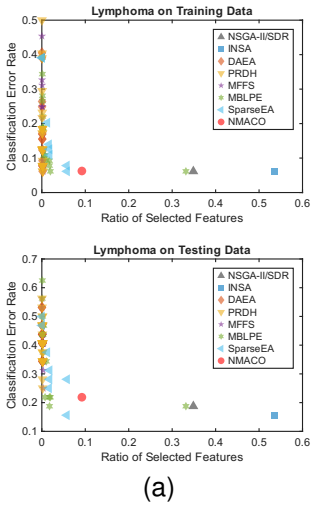


Fig. 6: Median Pareto fronts obtained by all compared algorithms on the remaining dataset (Part 4): (a) Lymphoma.